Modelling ice sheet hydrology

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Motivation

- Basal water controls ice sheet sliding and therefore ice discharge to the ocean.
- Ice-shelf hydrology can weaken buttressing and affect ice sheet stability.
- Periodic subglacial lake drainage can cause transient accelerations of ice flow.
- Freshwater discharge influences ocean circulation and ice-shelf melting.
- Hydrology is difficult to observe directly, making numerical models an essential tool.

Ice flow modelling

- Ice thickness evolves following mass conservation:

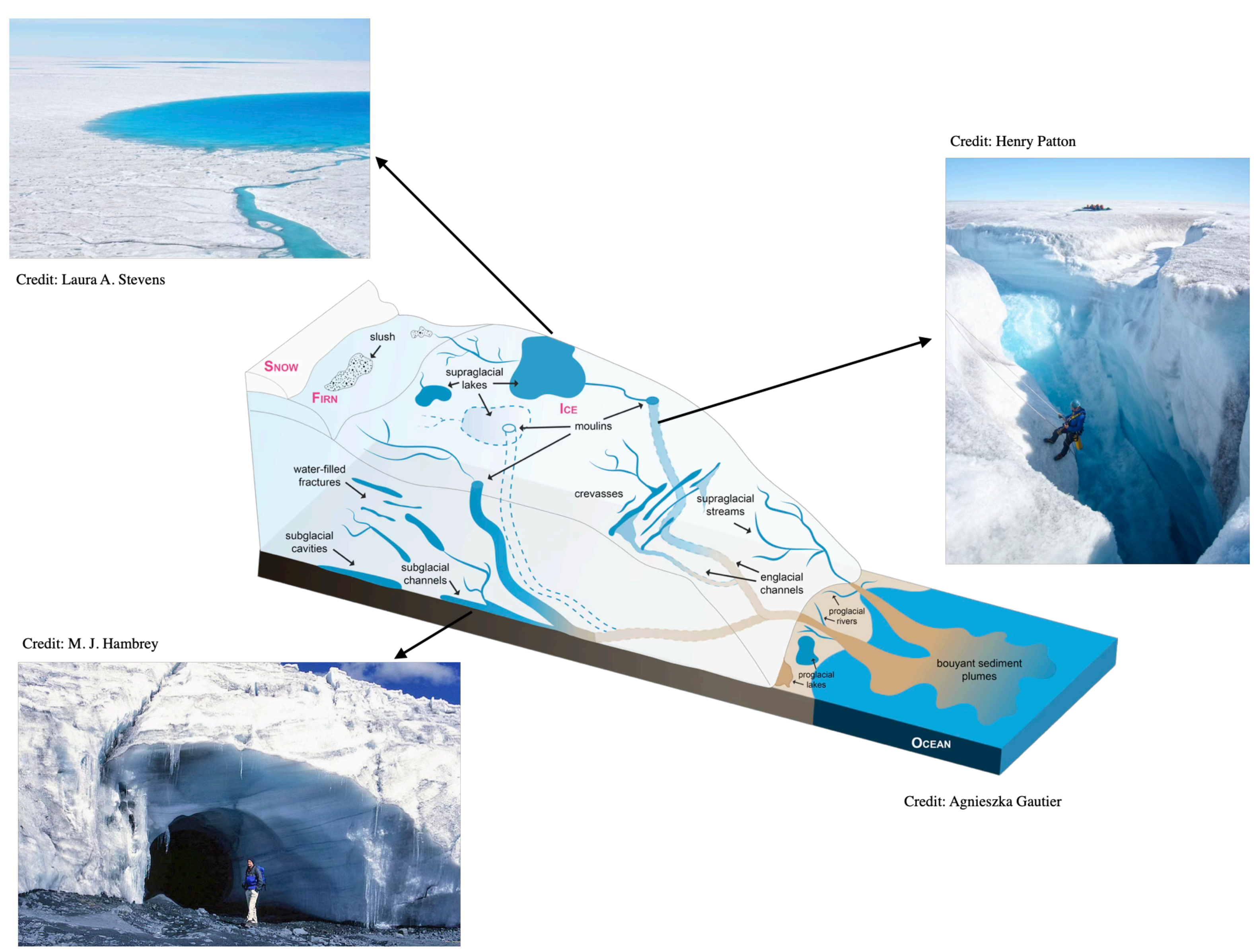
$$\frac{\partial H}{\partial t} = -\nabla \cdot (H\bar{\boldsymbol{u}}) + \dot{a}.$$

- A stress balance equation determines the ice velocity $\bar{\boldsymbol{u}}$.
- The basal shear stress $\boldsymbol{\tau}_{\boldsymbol{b}}(\bar{\boldsymbol{u}}, N)$ appears in this equation, which depends on the effective pressure $N = P_{\text{ice}} - P_{\text{water}}$.
- Subglacial hydrology determines P_{water}, thereby controlling basal resistance.
- An example of a sliding law is the regularized Coulomb law:

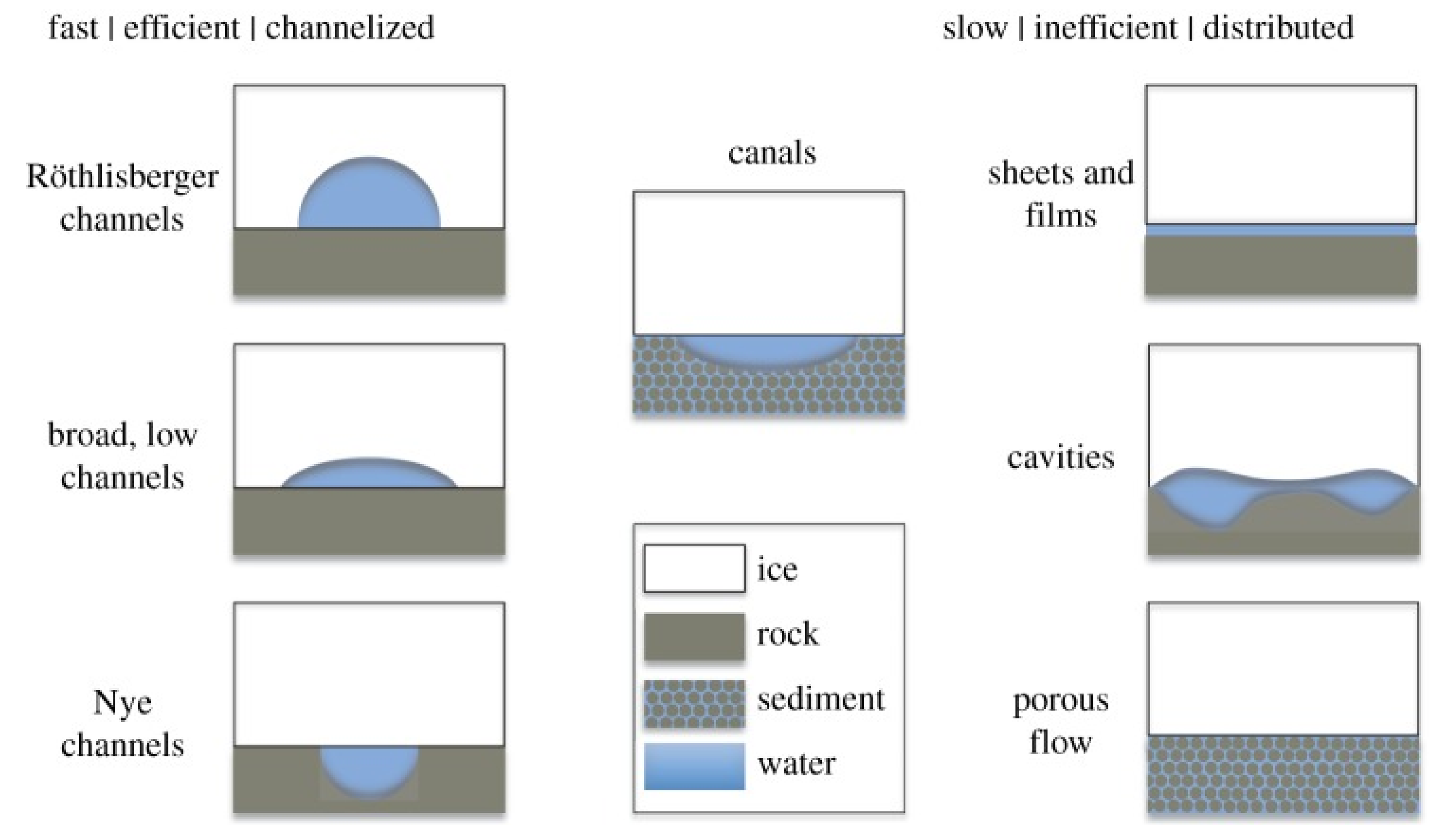
$$\boldsymbol{\tau}_b = C \frac{|\boldsymbol{u_b}|^{m-1}\boldsymbol{u_b}}{(|\boldsymbol{u_b}| + u_0)^m} N.$$

- Lower effective pressure (higher water pressure) reduces basal friction and increases ice flow speed.

Ice sheet hydrology



Ice sheet hydrology modelling



SHAKTI transient hydrology model

- Water conservation:

$$\frac{\partial b}{\partial t} = -\nabla \cdot \boldsymbol{q} + \frac{\dot{m}}{\rho_w} + i_{e \to b}.$$

- Cavity size also changes through melt opening, sliding over bed roughness, and viscous creep closure::

$$\frac{\partial b}{\partial t} = \frac{\dot{m}}{\rho_i} + \beta|\boldsymbol{u}_b| - A\,|N|^{n-1} N\, b.$$

- Basal melt rate $\dot{m}$ is generated by geothermal heating G, frictional heating from sliding, and dissipation of flowing water.
- Water flux is driven by gradients in hydraulic head $h = P_{\text{water}}/(\rho_w g) + z_{\text{bed}}$:

$$\boldsymbol{q} = \frac{b^3 g}{12\nu(1 + \omega Re)}(-\nabla h).$$

- We can solve for h, which gives P_{water}, and hence obtain N using $P_{\text{ice}} = \rho_i g H$.

Kori-ULB steady-state hydrology model

- Water conservation with steady state cavity size:

$$\nabla \cdot \boldsymbol{q} = \frac{\dot{m}}{\rho_w}.$$

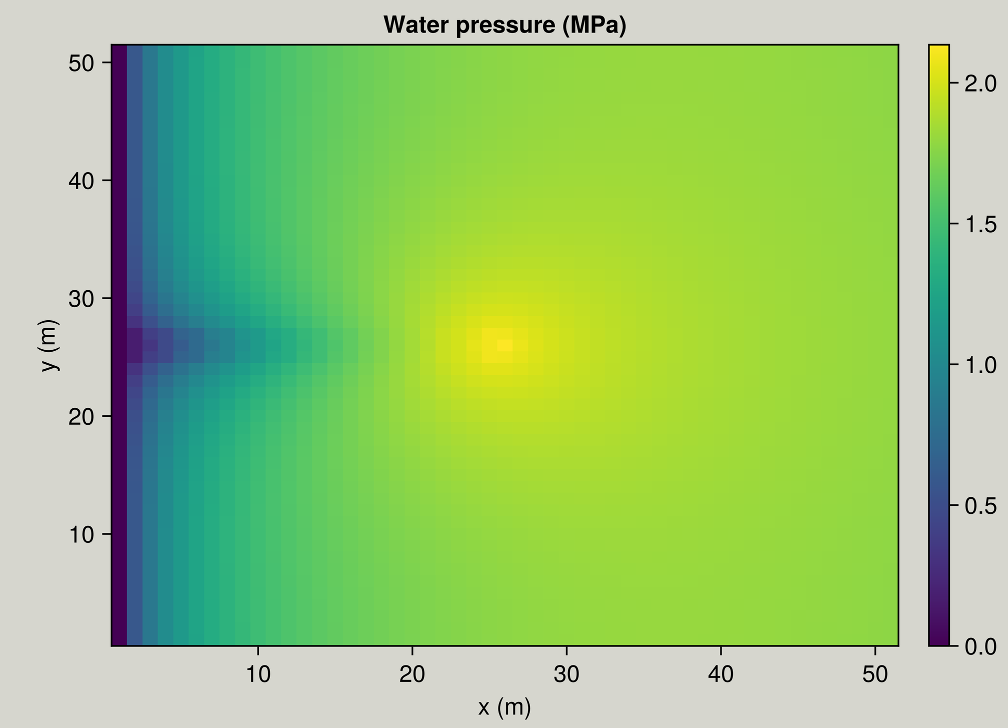
- Approximate the hydraulic potential by the geometric potential $\rho_w g \nabla h \approx \nabla \phi_0$.
- The conduit discharge is represented using a Darcy–Weisbach relation:

$$\boldsymbol{q} = K\, S^{5/4} |\nabla \phi_0|^{-1/2} (-\nabla \phi_0).$$

- Analytical expressions can then be obtained for conduit area S and effective pressure N.

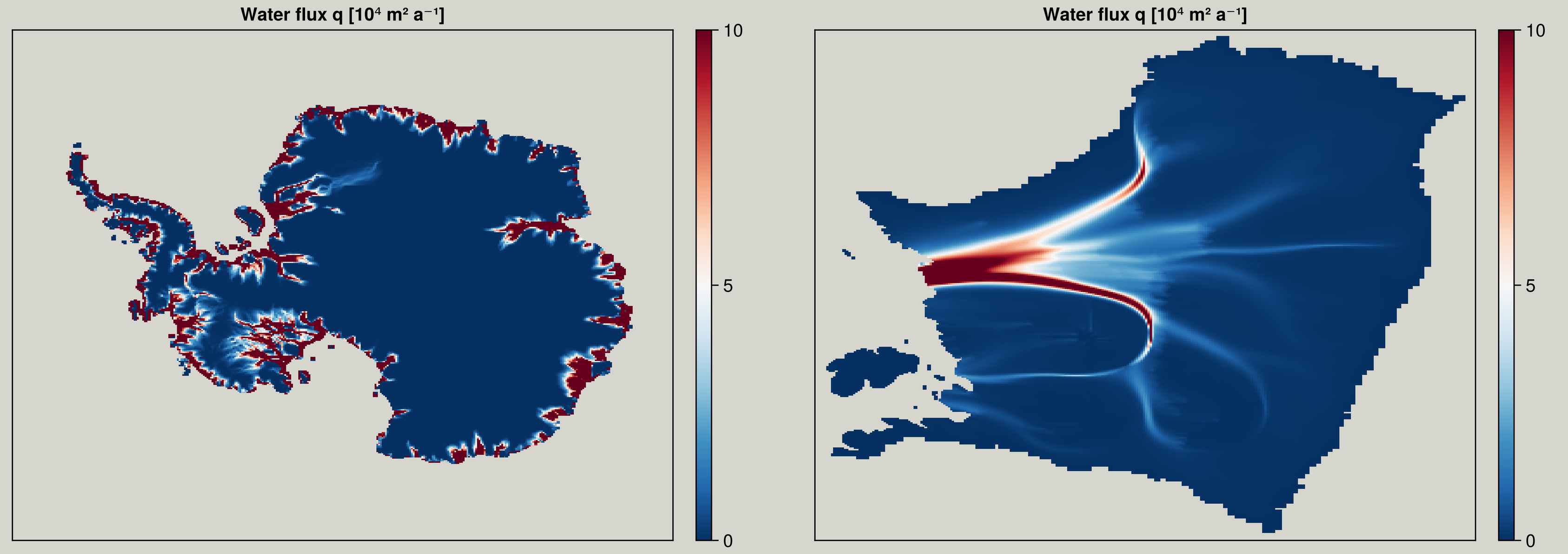
SHAKTI hydrology model results

- Water injected at the moulin initially spreads through the distributed drainage system.
- As discharge increases downstream, an efficient channel forms.
- The channel transmits large water fluxes at lower pressure and draws water from the surrounding distributed system.



Kori-ULB hydrology model results

- Water flux follows major ice streams and converges into efficient drainage pathways.
- The model reproduces large-scale drainage patterns across Antarctica.
- Beneath Thwaites Glacier, distributed flow transitions into channelized drainage near the grounding line.



Outlook

- Perform fully coupled ice sheet and hydrology simulations for Antarctica and Greenland.
- Assess the impact of hydrology on ice dynamics and sea-level projections.
- Investigate model sensitivity to spatial resolution, sliding law, and other parameters.
- Develop strategies for calibration and validation using observations.

References: [1] Flowers (2015), Proc. R. Soc. A, 471, 20140907, [2] Sommers et al. (2018), The Cryosphere, 12, 2955–2973, [3] Kazmierczak et al. (2024), The Cryosphere, 18, 5887–5911.